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Predicting Next Day Mean Temperature in Boston

Introduction

Boston experiences rapid weather changes and substantial temperature variation throughout the year. Atlantic influences, continental air masses, and the nearby jet stream contribute to the region's variable conditions (Runkle et al., 2022). At Boston Logan International Airport, the 1991–2020 normal monthly mean temperature ranges from approximately −1.2°C in January to 23.4°C in July (National Centers for Environmental Information [NCEI], n.d.). These seasonal differences and short-term fluctuations motivate examining how much information recent local weather provides about tomorrow's temperature. Short-term temperature estimates can inform everyday planning and heating and cooling management (Wang et al., 2021).

Previous research has explored statistical and machine learning approaches to temperature prediction. Wang et al. (2021) used daily temperature observations from Alpena, Michigan, and reported that triple exponential smoothing outperformed a persistence baseline, which assumes temperature remains unchanged: tomorrow's predicted mean temperature equals today's mean temperature. Their study supports including persistence as a meaningful benchmark. Barnor et al. (2026) compared 17 models across 65 locations. They found that machine learning models outperformed benchmark models at forecasting horizons relevant to weather derivatives. A forecasting horizon is how far into the future a prediction extends. Although their focus included longer forecasting windows for weather derivatives, their findings motivate evaluating competing models under a common experimental design. Some studies also use ERA5 data to predict weather conditions. For example, Rasp et al. (2020) used ERA5 data to develop WeatherBench, a benchmark for forecasting weather three and five days ahead. They compared persistence, climatology, linear regression, a convolutional neural network, and physical

numerical weather prediction (NWP) models. For two-meter temperature, their results showed that the direct linear regression model had a lower RMSE than the persistence baseline for both three-day and five-day forecasts.

In this project, we predict the next day's mean temperature in Boston using only the temperature, precipitation, wind speed and snowfall of the previous three days. We use daily ERA5 reanalysis data from 2006 to 2025 (Hersbach et al., 2020), obtained through the Open-Meteo Historical Weather API (Zippenfenig, 2024). We test whether regression models using a few recent local variables can predict tomorrow's temperature more accurately than simply assuming it will equal today's.

Data

We collected our data from the Open-Meteo Historical Weather API (Zippenfenig, 2024), which serves ERA5 reanalysis data (Hersbach et al., 2020). A Python script queried daily weather for Boston (42.361°N, 71.057°W) from 1 January 2006 to 31 December 2025, using the America/New_York time zone. The API returns the nearest ERA5 grid cell (42.25°N, 71.00°W, elevation 9 m), so values describe a cell of about 25 km around Boston rather than a single station. We stored the raw JSON response, including units, and the daily table.

The raw dataset contains 7,305 daily records and five columns: date, daily mean temperature (°C), total precipitation (mm), maximum wind speed (km/h), and snowfall (cm). We found no duplicated dates, calendar gaps or missing values. Because we predict tomorrow's temperature, we turned the daily table into a supervised dataset. The target is the daily mean temperature on day t, the average of the hourly air temperature at 2 m above ground. In the training years (2006–2017), it has a mean of 10.94 °C and a standard deviation of 9.11 °C, and ranges from −17.4 to 30.6 °C. For each target day t, the features are the four weather variables from days t-1, t-2 and t-3, giving 12 continuous features. The first three days of 2006 lack a full history, leaving 7,302 samples. All features are continuous, and the dataset contains no categorical or ordinal columns.

As observations are ordered in time and not i.i.d., we split by target date rather than randomly: training 2006–2017 (4,380 rows, 60%), validation 2018–2021 (1,461 rows, 20%) and test 2022–2025 (1,461 rows, 20%). Detailed column descriptions are in the GitHub repository.

Exploratory Data Analysis

We explored the data in three steps. First, we checked data quality on the raw dataset. Then, we examined the distribution of each variable. Finally, we studied how the variables change with the seasons and with each other. All explorations and plots use only the training period (2006–2017), so that no information from the validation and test years influences our modeling decisions. The three plots below carry the main messages for our prediction task.

Figure 1 displays the seasonal cycle of temperature and precipitation. Mean daily temperature rises from −0.5 °C in January to 22.8 °C in July. Mean daily precipitation, in contrast, stays between 2.5 mm (September) and 3.7 mm (December) in every month. We found temperature follows a strong yearly cycle, while precipitation is spread across the year. Although we do not include calendar features, recent weather conditions may indirectly provide information about the season.

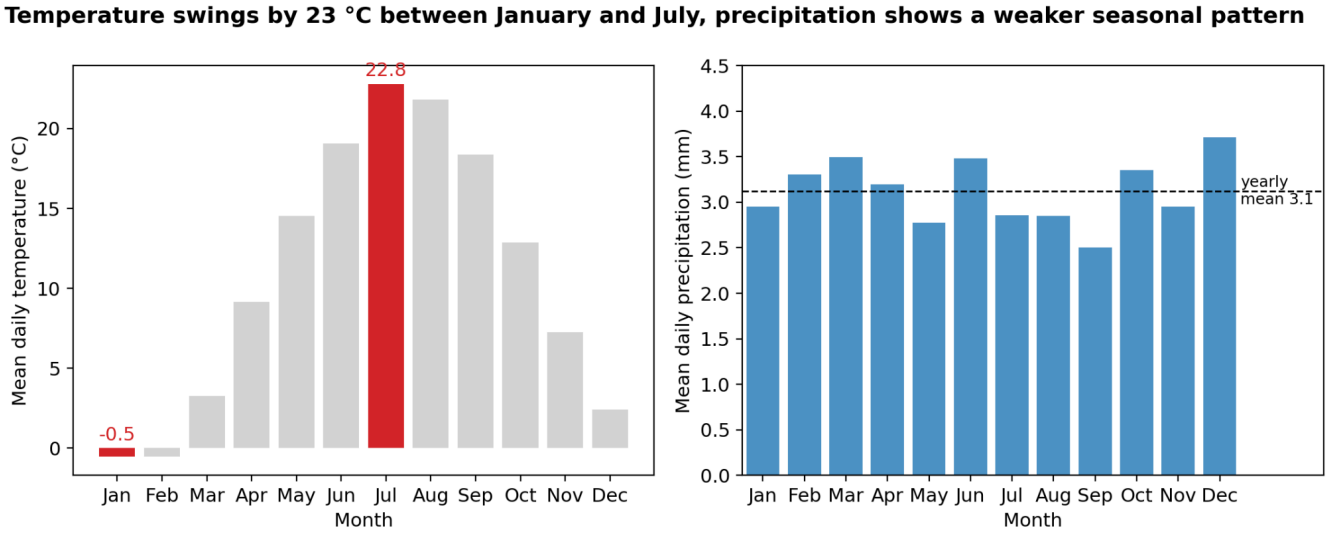


Figure 1. Mean daily temperature (left) and mean daily precipitation (right) by month in the training period (2006–2017). The dashed line marks the yearly mean precipitation.

Figure 2 shows that wind also follows the seasons. The mean daily maximum wind speed is highest in February (23.9 km/h) and lowest in August (16.4 km/h), about 46% higher. The typical level of wind speed thus depends on the time of year as well. Wind speed may therefore also help the model recognize the season.

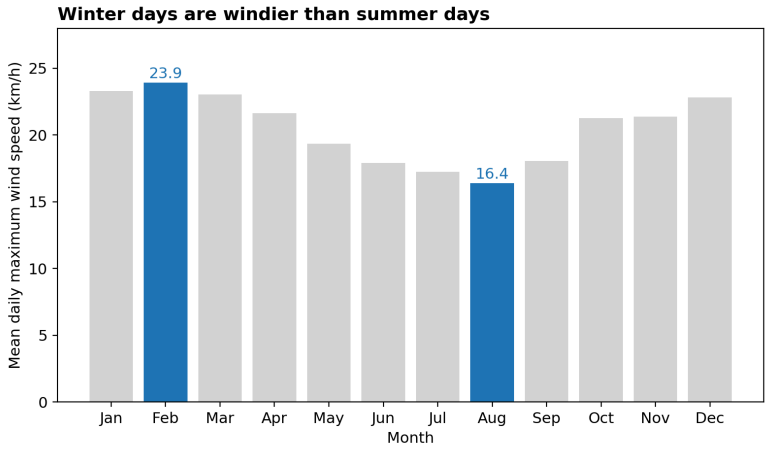


Figure 2. Mean daily maximum wind speed by month in the training period (2006–2017). The windiest and calmest months are highlighted.

Figure 3 compares the temperature on all days with the temperature on the 438 days with snowfall in the training dataset. Snowy days are concentrated around 0 °C, and 95% of them are colder than 5 °C, while only about 30% of all days are that cold. Snowfall and temperature are therefore closely linked, so these features may carry overlapping information.

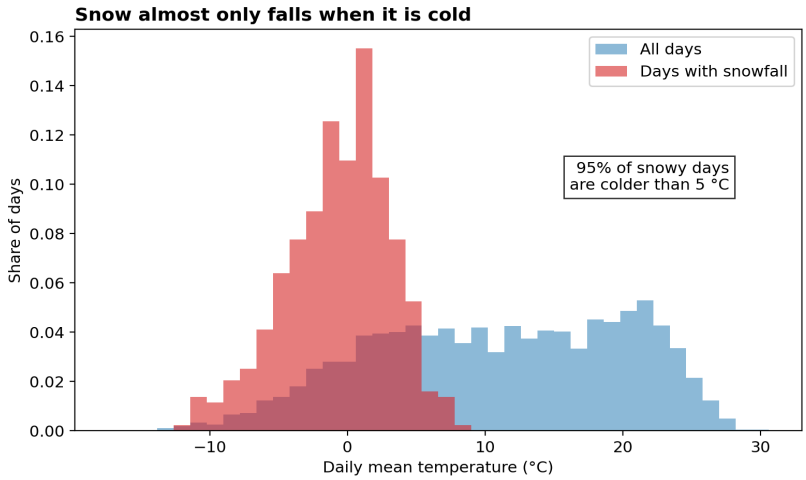


Figure 3. Distribution of daily mean temperature on all days and on days with snowfall in the training period (2006–2017). Each histogram shows the share of days per temperature bin.

Problem Description and Potential Impact

This project aims to predict the next day's mean temperature in Boston using weather conditions from the preceding three days. The target variable is the daily mean temperature of the next day, measured in degrees Celsius (°C). This is a regression problem, because temperature is measured on a continuous scale.

To address this problem, we plan to compare the performance of several models, including Linear Regression, Ridge Regression, and Random Forest, to investigate which model has the best prediction accuracy. We will also compare them with a persistence baseline, which predicts that tomorrow's mean temperature will be the same as today's temperature. We will evaluate model performance mainly using mean absolute error (MAE), as it is the most straightforward way that shows the average absolute difference between predicted and actual temperatures. We will also use the root mean square error (RMSE) because it is more sensitive to large prediction errors.

This project might help families and managers of rental properties plan the use of heating and cooling equipment in advance based on the next day's mean temperature. We believe these predictions could be useful for helping them maintain comfortable indoor temperatures, especially during severe winter weather in Boston. Our project uses historical ERA5 data (Hersbach et al., 2020), which have a publication delay of about five days. To use our model for real-time prediction, we could obtain recent weather inputs through Open-Meteo's Forecast API using its past_days feature (Open-Meteo, n.d.). Since these inputs are different from the ERA5 data we used for training, which may affect prediction accuracy, the model would need further testing before deployment. If any features are still unavailable at prediction time, we would drop those features and retrain the model.

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